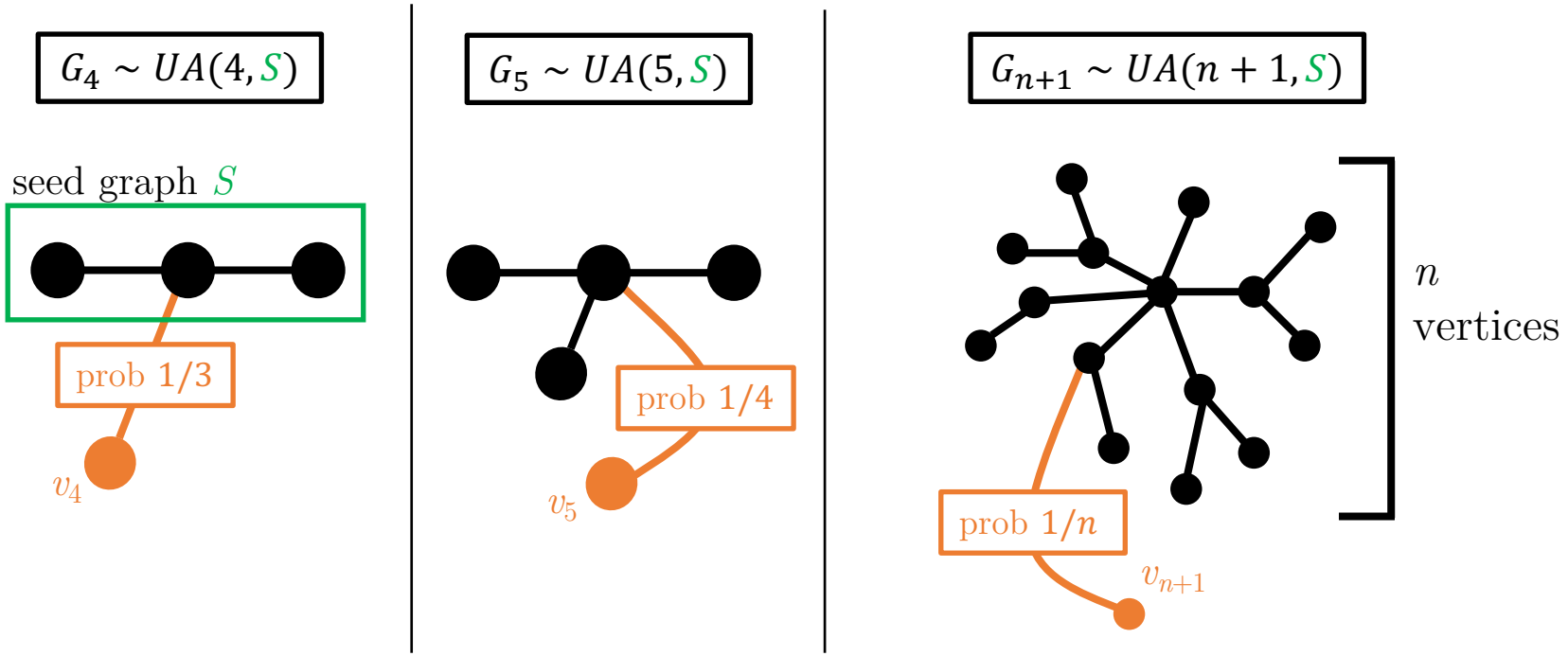
# Part I: Correlated Randomly Growing Graphs

**Joint work with Miklós Rácz**

# Randomly growing graphs

**Example:** Uniform Attachment (UA) trees

*Growth rule:* New vertices attach to a uniform random existing vertex.




In Preferential Attachment (PA), probability of connecting to an existing vertex is proportional to the degree of the vertex.

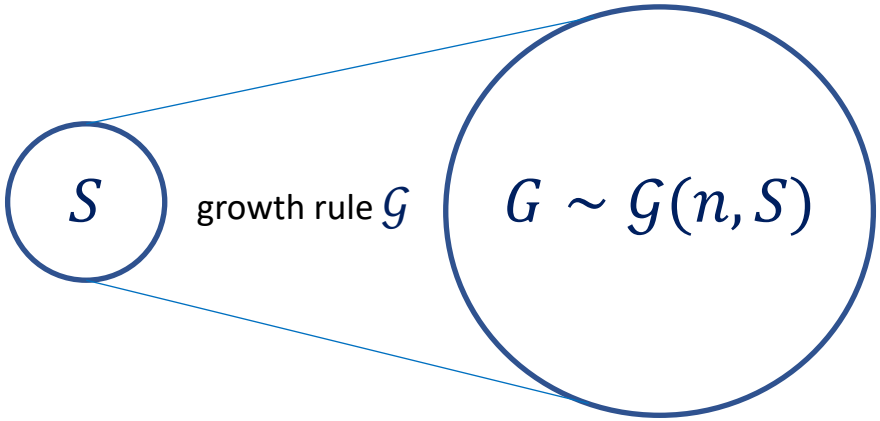
# Correlated growing graphs

Models of randomly growing graphs:

- Start from a seed graph $S$
- At each time step, add a new vertex with some new edges

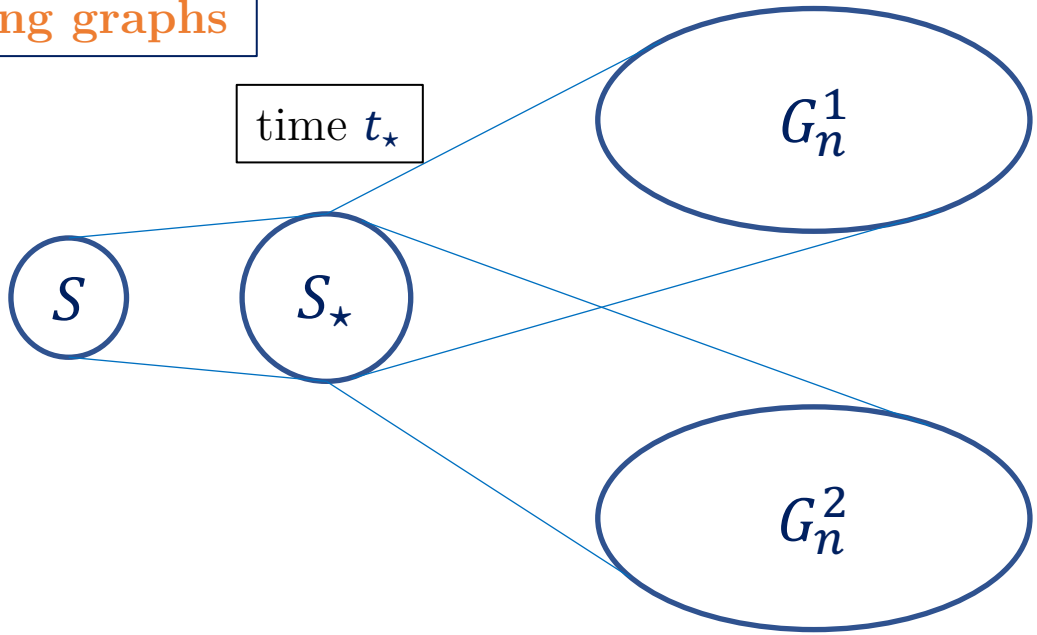
Canonical examples:

- uniform attachment (UA)
- preferential attachment (PA)



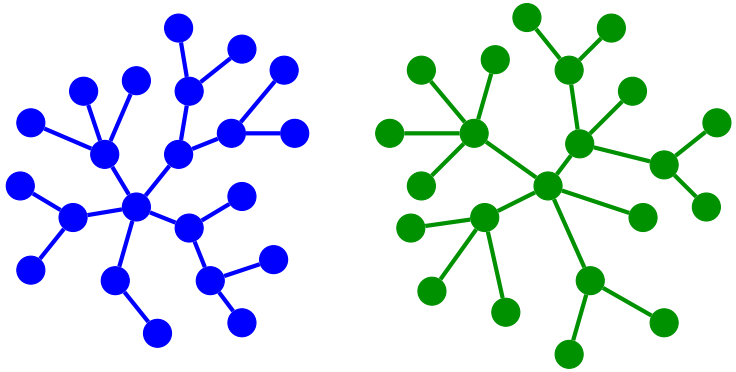

**New model: correlated growing graphs**

- Start from seed graph $S$
- Grow together until time $t_\star$:
  - $S_\star \sim \mathcal{G}(t_\star, S)$
- Then grow independently:
  - $G_n^1 \sim \mathcal{G}(n, S_\star)$
  - $G_n^2 \sim \mathcal{G}(n, S_\star)$
- $(G_n^1, G_n^2) \sim \mathcal{CG}(n, t_\star, S)$



# Correlated growing graphs

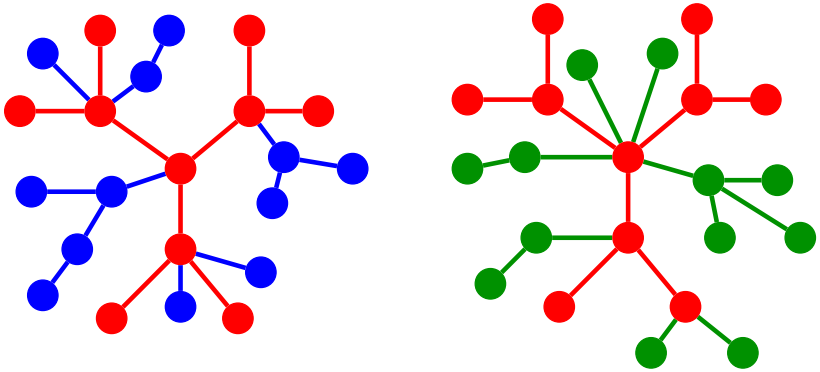
**Independent graphs**



$(G_n^1, G_n^2) \sim \mathcal{G}(n, S)^{\otimes 2}$

**Correlated graphs**

$S_{t_\star} \sim \mathcal{G}(t_\star, S)$



$(G_n^1, G_n^2) \sim \mathcal{CG}(n, t_\star, S)$

**Examples of correlated structure in growing graphs:**

- Initial user base of two social networks were the same, and then the social networks evolved independently
- Citation networks of two fields that shared common roots but grew apart
- A virus mutates into two different strains when propagating on a network

**Q:** Given $G_n^1, G_n^2$, can we **detect** correlation? Can we **estimate** $t_\star$ and $S_{t_\star}$?

# Related work: Graph matching

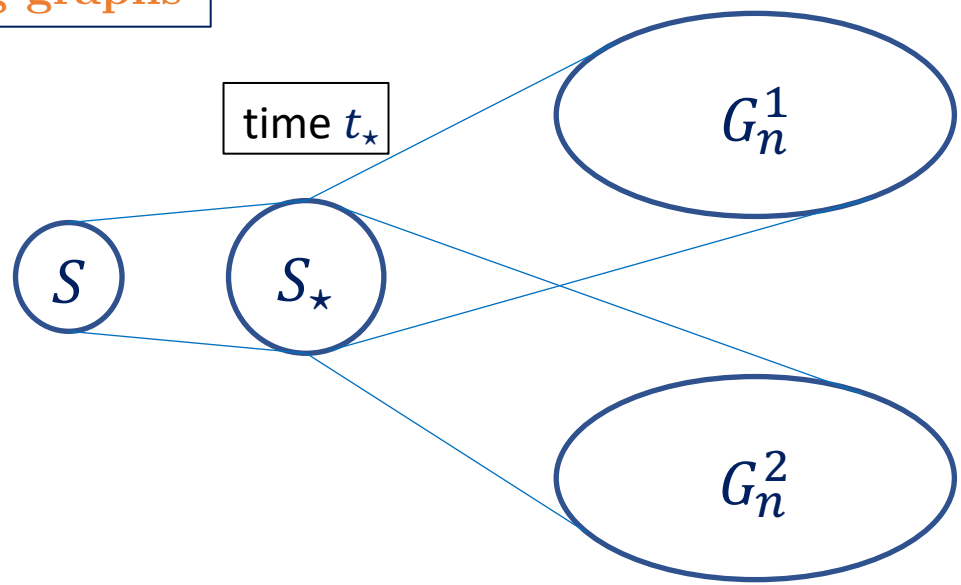
- **Goal:** given $G_1$ and $G_2$, find a matching of the vertices that *minimizes* some notion of *dissimilarity*. Ex: # edges present in one graph, but not the other.
- **Applications:**
  - Computer vision (Cho, Lee, 2012)
  - Pattern recognition (Conte et al., 2004)
  - Social networks (Narayanan, Shmatikov, 2008)
  - Computational biology (Singh, Xu, Berger, 2008)
- **Hardness:** NP-hard, even to approximate
- **Probabilistic generative models:**
  - Correlated Erdős-Rényi random graphs: Pedersani, Grossglauser, 2011
  - Many works in probability/statistics/CS/info theory… including:
  - Cullina, Kiyavash, 2016, 2018
  - Barak, Chou, Lei, Schramm, Sheng, 2018
  - Ding, Ma, Wu, Xu, 2018
  - Mossel, Xu, 2019
  - Fan, Mao, Wu, Xu, 2019, 2019
  - Correlated preferential attachment graphs: Korula, Lattanzi, 2014

Fundamental info-theoretic limits and algorithmic advances

# Detecting correlation

## Correlated randomly growing graphs

- Start from seed graph $S$
- Grow together until time $t_\star$:
  - $S_\star \sim \mathcal{G}(t_\star, S)$
- Then grow independently:
  - $G_n^1 \sim \mathcal{G}(n, S_\star)$
  - $G_n^2 \sim \mathcal{G}(n, S_\star)$
- $(G_n^1, G_n^2) \sim \mathcal{CG}(n, t_\star, S)$




## Detection:

$$H_0 : \left(G_n^1, G_n^2\right) \sim \mathcal{G}\left(n, S\right)^{\otimes 2} \qquad H_1 : \left(G_n^1, G_n^2\right) \sim \mathcal{CG}\left(n, t_\star, S\right)$$

independent graphs — correlated graphs

**Q:** Does there exist a hypothesis test with positive power?

Object of study: $\lim_{n\to\infty} \mathrm{TV}\left(\mathcal{CG}\left(n, t_\star, S\right), \mathcal{G}\left(n, S\right)^{\otimes 2}\right)$

# Seed has influence ⇒ detection

**Theorem: Detection with positive power**

Fix a seed graph $S$, a positive integer $t_\star$ such that $\boxed{t_\star > |S|}$, and a Markov attachment rule $\mathcal{G}$.
Suppose that there are graphs $G$ and $G'$ on $t_\star$ vertices such that $\mathbb{P}_S(G) > 0$ and $\mathbb{P}_S(G') > 0$, and that

$$\boxed{\delta_{\mathcal{G}}(G, G') := \lim_{n\to\infty} \mathrm{TV}\left(\mathcal{G}(n, G), \mathcal{G}(n, G')\right) > 0}$$

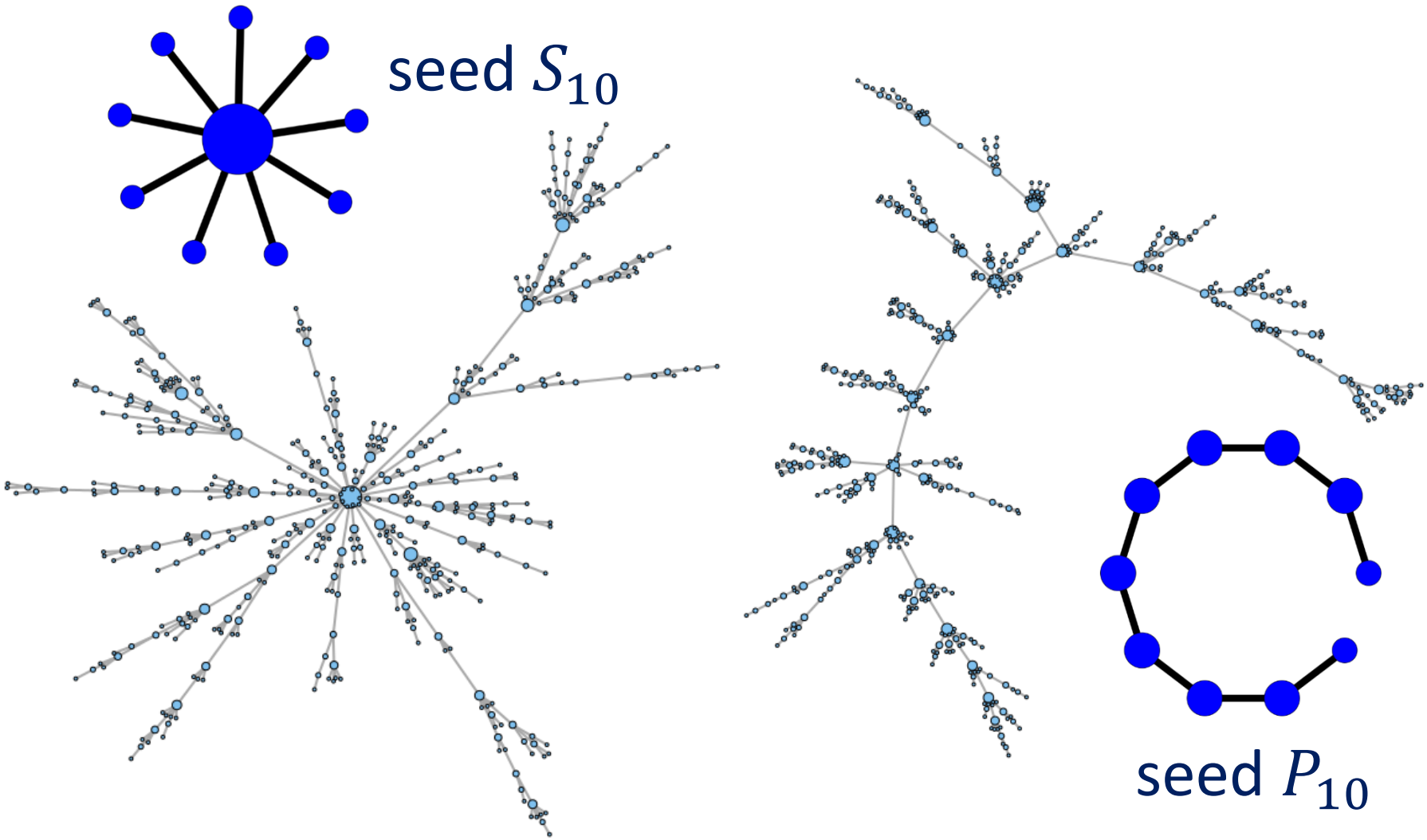
Then:

seed has influence

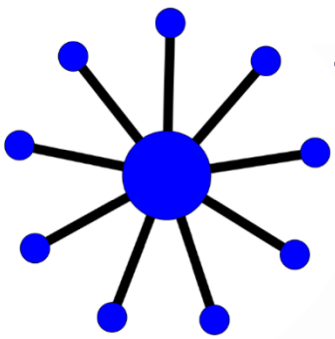
$$\lim_{n\to\infty} \mathrm{TV}\left(\mathcal{CG}(n, t_\star, S), \mathcal{G}(n, S)^{\otimes 2}\right) > 0.$$

Detection possible even if graphs are grown together for a **single time step!**

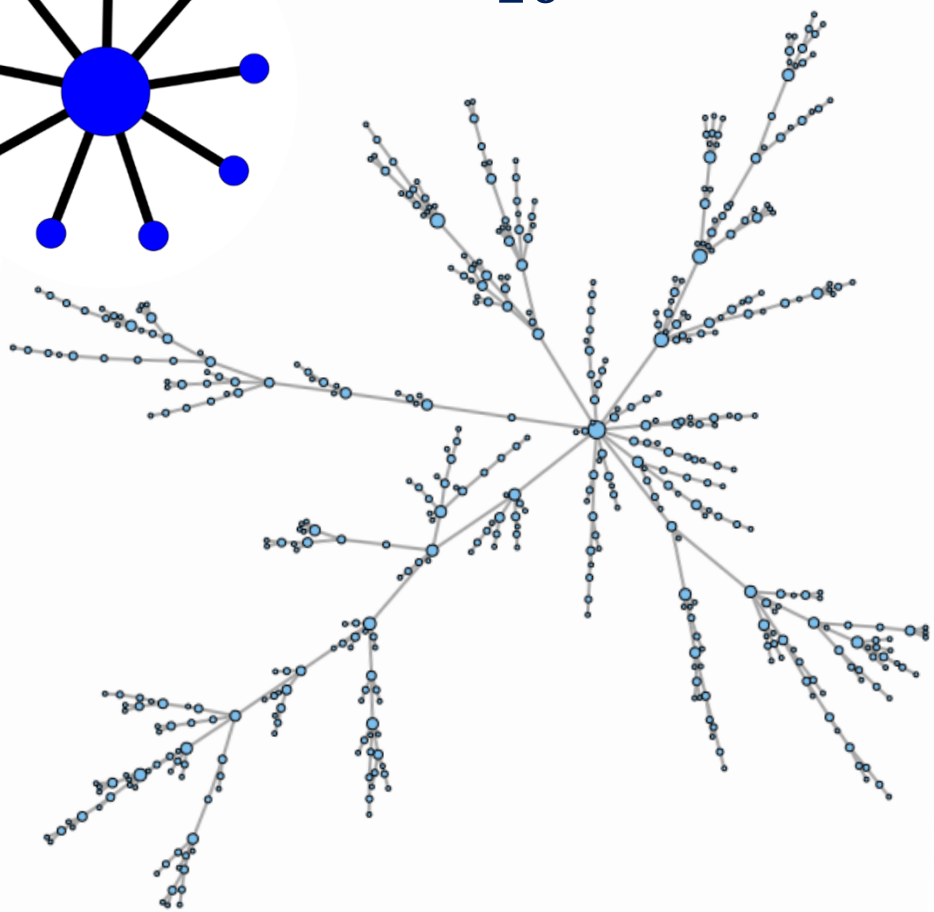
# Influence of the seed: PA trees

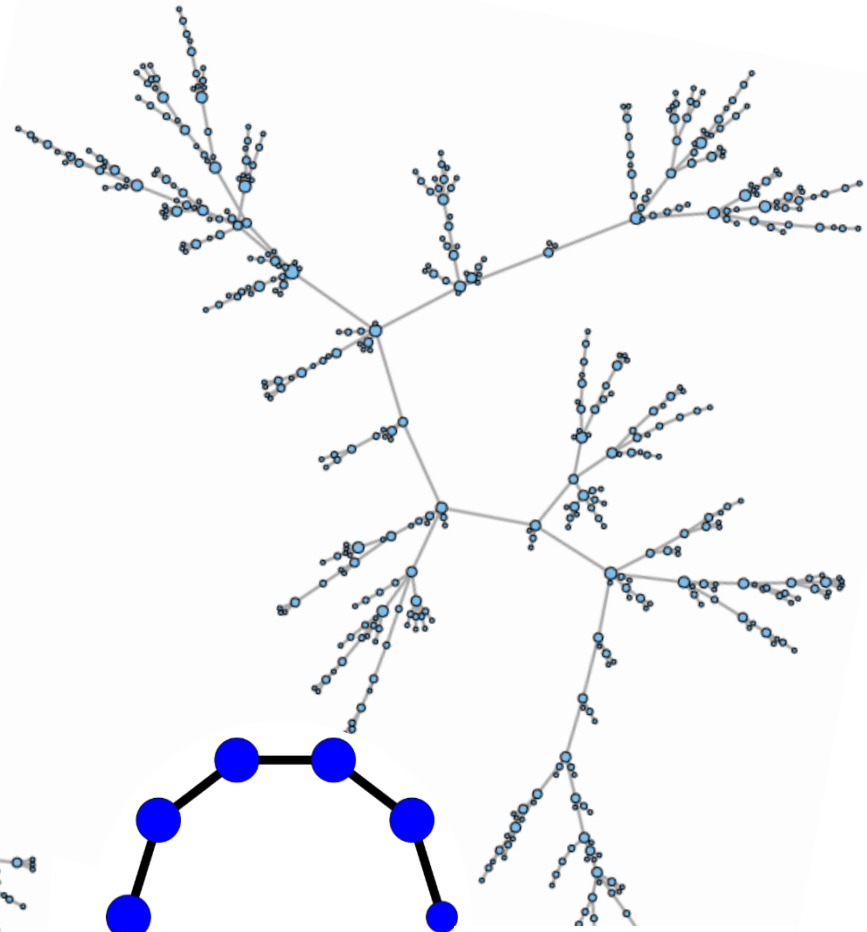


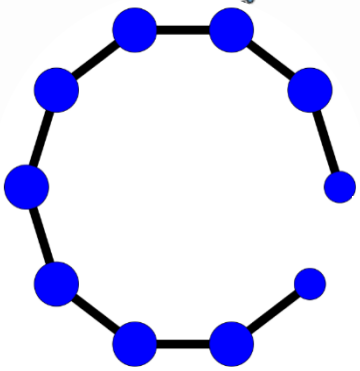
# Influence of the seed: UA trees



seed $S_{10}$







seed $P_{10}$

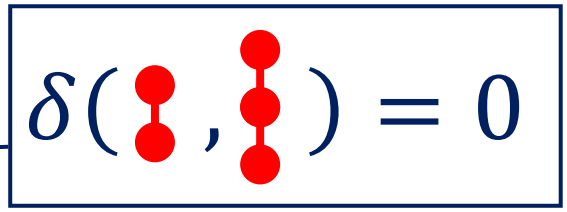
# Influence of the seed

Preferential attachment:

**Thm (Bubeck, Mossel, Rácz, 2015)**

If the degree profiles of $S$ and $T$ are different, and both have at least 3 vertices, then

$$\delta_{\mathrm{PA}}(S, T) > 0.$$



**Thm (Curien, Duquesne, Kortchemski, Manolescu, 2015)**

If $S$ and $T$ are non-isomorphic, and both have at least 3 vertices, then

$$\delta_{\mathrm{PA}}(S, T) > 0.$$

Uniform attachment:

**Thm (Bubeck, Eldan, Mossel, Rácz, 2017)**

If $S$ and $T$ are non-isomorphic, and both have at least 3 vertices, then

$$\delta_{\mathrm{UA}}(S, T) > 0.$$

# Seed has influence ⇒ detection

**Theorem: Detection with positive power**

Fix a seed graph $S$, a positive integer $t_\star$ such that $t_\star > |S|$, and a Markov attachment rule $\mathcal{G}$.
Suppose that there are graphs $G$ and $G'$ on $t_\star$ vertices such that $\mathbb{P}_S(G) > 0$ and $\mathbb{P}_S(G') > 0$, and that

influence of the seed

$$\boxed{\delta_{\mathcal{G}}(G, G') := \lim_{n\to\infty} \mathrm{TV}\left(\mathcal{G}(n, G), \mathcal{G}(n, G')\right) > 0}\ .$$

Then:

$$\lim_{n\to\infty} \mathrm{TV}\left(\mathcal{CG}(n, t_\star, S), \mathcal{G}(n, S)^{\otimes 2}\right) > 0\ .$$

**Corollary**

Let $S$ be a finite tree with at least two vertices, and let $t_\star$ be such that $t_\star > |S|$ and $t_\star > 3$. Then:

$$\lim_{n\to\infty} \mathrm{TV}\left(\mathrm{CPA}(n, t_\star, S), \mathrm{PA}(n, S)^{\otimes 2}\right) > 0$$

$$\lim_{n\to\infty} \mathrm{TV}\left(\mathrm{CUA}(n, t_\star, S), \mathrm{UA}(n, S)^{\otimes 2}\right) > 0$$

"Existence" proof

# Detection in PA trees

**Corollary**

Let $S$ be a finite tree with at least two vertices, and let $t_\star$ be such that $t_\star > |S|$ and $t_\star > 3$. Then:

$$\lim_{n\to\infty} \mathrm{TV}\left(\mathrm{CPA}\left(n, t_\star, S\right), \mathrm{PA}\left(n, S\right)^{\otimes 2}\right) > 0.$$

$$A_{u,n} := \left\{ \boxed{\frac{\Delta\left(T_1\left(n\right)\right)}{\sqrt{n}}} > u, \boxed{\frac{\Delta\left(T_2\left(n\right)\right)}{\sqrt{n}}} > u \right\}$$

normalized max degree

$$\lim_{n\to\infty} \mathbb{P}_0\left(A_{u,n}\right) \sim \frac{16}{\pi} u^{-2} \exp\left(-u^2/2\right)$$

For the seed $S_2$

$$\lim_{n\to\infty} \mathbb{P}_{t_\star}\left(A_{u,n}\right) \sim \frac{2^{t_\star+1}}{\pi} u^{-2} \exp\left(-u^2/2\right)$$

# Detection in UA trees

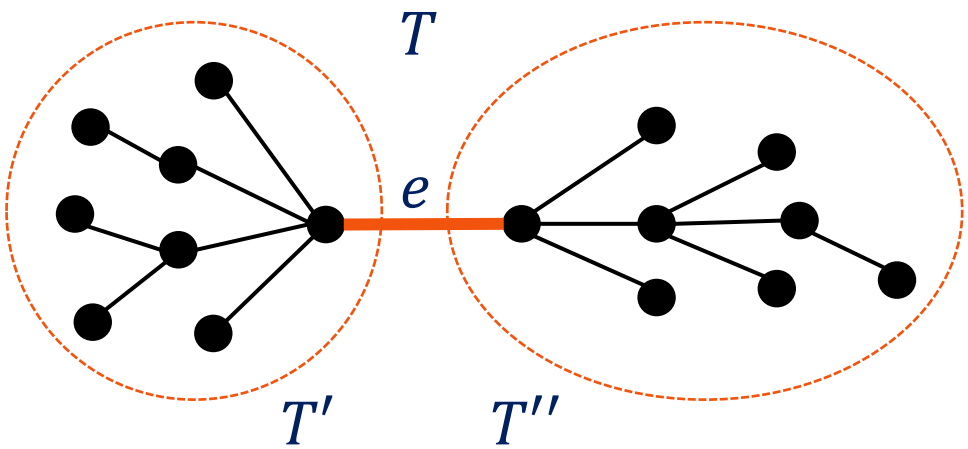
**Corollary**

Let $S$ be a finite tree, and let $t_\star$ be such that $t_\star > |S|$ and $t_\star > 3$. Then:

$$\lim_{n\to\infty} \mathrm{TV}\left(\mathrm{CUA}\left(n, t_\star, S\right), \mathrm{UA}\left(n, S\right)^{\otimes 2}\right) > 0.$$

$$h(T,e) := \frac{|T'|^2|T''|^2}{|T|^4}$$

$$H(T) := \sum_{e\in E(T)} h(T,e)$$




$H(T_n^1)H(T_n^2)$ or equivalently $\left(H(T_n^1) - H(T_n^2)\right)^2$ detects correlation

# Detecting correlation w.h.p.

**Detection:**

$$H_0 : \left(G_n^1, G_n^2\right) \sim \mathcal{G}\left(n, S\right)^{\otimes 2} \qquad H_1 : \left(G_n^1, G_n^2\right) \sim \mathcal{CG}\left(n, t_\star, S\right)$$

independent graphs — correlated graphs

Ideally: $$\mathrm{TV}\left(\mathcal{CG}\left(n, t_\star, S\right), \mathcal{G}\left(n, S\right)^{\otimes 2}\right) \approx 1$$

However, for any fixed $t_\star$, there exists $\varepsilon = \varepsilon(\mathcal{G}, t_\star) > 0$, such that for every $n \geq t_\star$:

$$\mathrm{TV}\left(\mathcal{CG}\left(n, t_\star, S\right), \mathcal{G}\left(n, S\right)^{\otimes 2}\right) \leq 1 - \varepsilon$$

# Detection correlation w.h.p.

**Theorem (Rácz, Sridhar, 2020)**

Fix a seed graph $S$, and a Markov attachment rule $\mathcal{G}$. Let $\{G_n\}_{n \geq |S|}$ be a sequence of randomly growing graphs with seed $S$ and attachment rule $\mathcal{G}$.

Suppose that there exists a function $f$ such that $\lim_{n\to\infty} f(G_n) =: f_\infty$ exists almost surely and $f_\infty$ is an absolutely continuous random variable.

Then: $$\lim_{t_\star \to \infty} \lim_{n \to \infty} \mathrm{TV}\left(\mathcal{CG}\left(n, t_\star, S\right), \mathcal{G}\left(n, S\right)^{\otimes 2}\right) = 1$$

Proof idea: consider the statistic $\left| f\left(G_n^1\right) - f\left(G_n^2\right) \right|$

For large $t_\star$ and an appropriately chosen $\varepsilon$, we have that

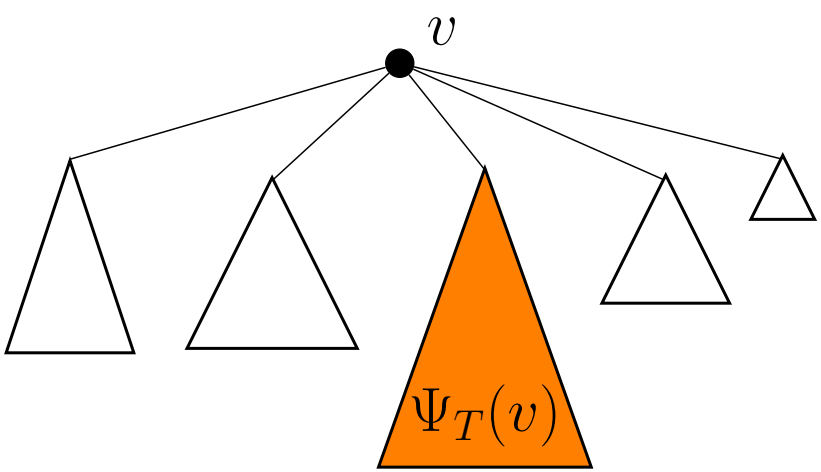
$$\mathbb{P}_0\left(\left| f\left(G_n^1\right) - f\left(G_n^2\right) \right| \leq \varepsilon\right) \approx 0 \qquad \mathbb{P}_{t_\star}\left(\left| f\left(G_n^1\right) - f\left(G_n^2\right) \right| \leq \varepsilon\right) \approx 1$$

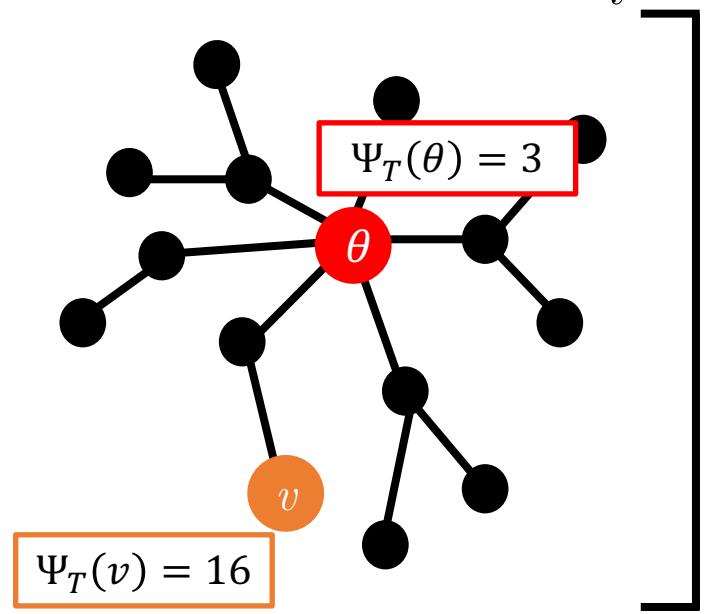
# Centrality in trees

Anti-centrality of $v$: $\Psi_T(v) := \max_{u \in \mathcal{N}_v(T)} |(T, v)_{u\downarrow}|$

The *centroid* $\theta$ is the vertex with minimum anti-centrality.

$\Psi_T(v)$ and variants have been well-studied[1,2,3]




17 nodes

- For a fixed $v$, $\lim_{n\to\infty} \frac{1}{n} \Psi_{T_n}(v)$ exists a.s. and is absolutely continuous
- The centroid $\theta_n$ of $T_n$ is *persistent* in UA and PA trees[4]

$$\lim_{n\to\infty} \frac{1}{n} \Psi_{T_n}(\theta_n) = \lim_{n\to\infty} \frac{1}{n} \min_{v \in T_n} \max_{u \in \mathcal{N}_v(T_n)} |(T_n, v)_{u\downarrow}|$$

exists a.s. and is absolutely continuous


1. D. Shah and T. Zaman. Detecting Sources of Computer Viruses in Networks: Theory and Experiment. In ACM SIGMETRICS, volume 38, pages 203–214, 2010.
2. D. Shah and T. Zaman. Rumors in a Network: Who's the Culprit? IEEE Transactions on Information Theory, 57(8):5163–5181, 2011.
3. S. Bubeck, L. Devroye, and G. Lugosi. Finding Adam in random growing trees. Random Structures & Algorithms, 50(2):158–172, 2017.
4. V. Jog and P.-L. Loh. Persistence of centrality in random growing trees. Random Structures & Algorithms, 52(1):136–157, 2018.

# Estimating $t_\star$

**Theorem (Rácz, Sridhar, 2020)**

Let $(T_n^1, T_n^2) \sim CPA(n, t_\star, S)$. There exists an estimator $\hat{t} = \hat{t}(T_n^1, T_n^2)$ computable in polynomial time such that

$$\lim_{t_\star \to \infty} \liminf_{n \to \infty} \mathbb{P}\left(\left(1 - \frac{\log\log t_\star}{\sqrt{\log t_\star}}\right) t_\star \leq \widehat{t}_n \leq \left(1 + \frac{\log\log t_\star}{\sqrt{\log t_\star}}\right) t_\star\right) = 1.$$

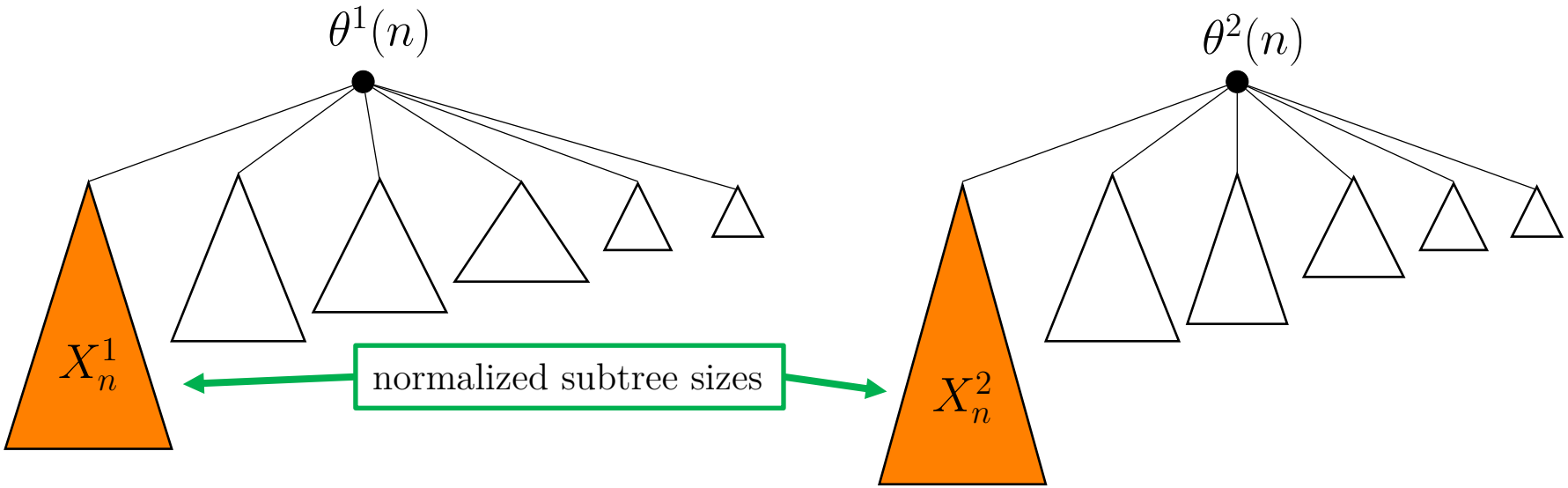
The same holds for UA trees. .

In words:

we can estimate $t_\star$ with *vanishing relative error* as $t_\star \to \infty$.

# Proof ideas

Matching the centroids and their subtrees

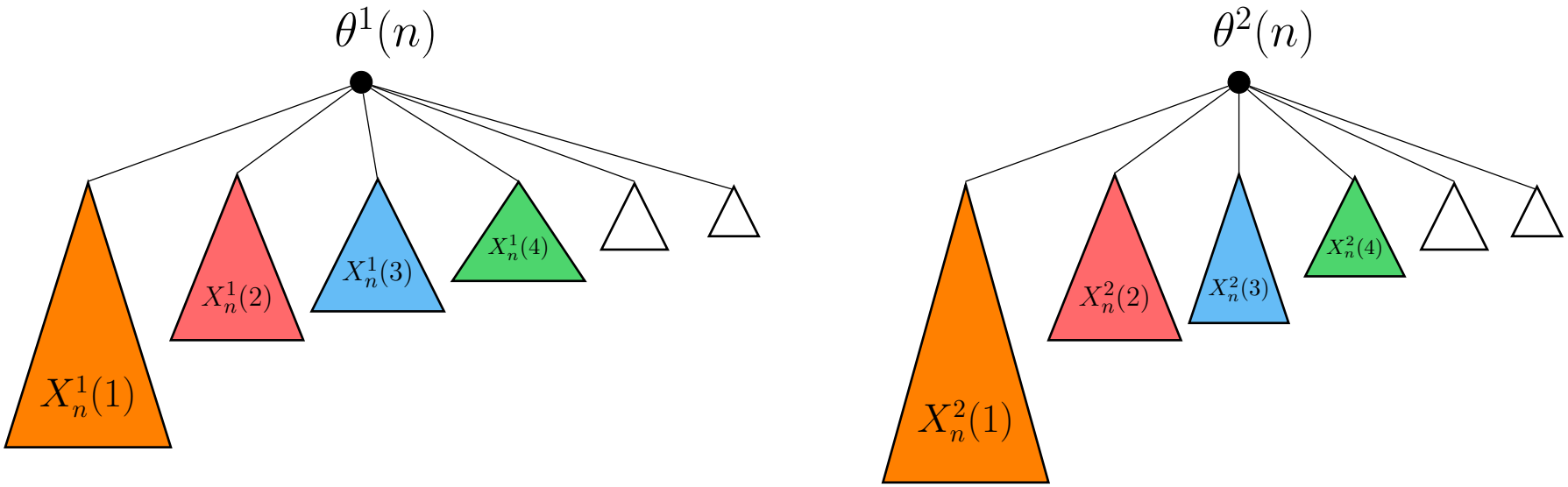



- Idea behind estimator: $(X_n^1 - X_n^2)^2 \asymp \frac{1}{t_\star}$
- **Coarse estimate** obtained by matching largest subtrees of the centroid

$$\boxed{\frac{t_\star}{\log t_\star} \leq \hat{t} \leq t_\star \log t_\star}$$

# Proof ideas

Matching the centroids and their subtrees



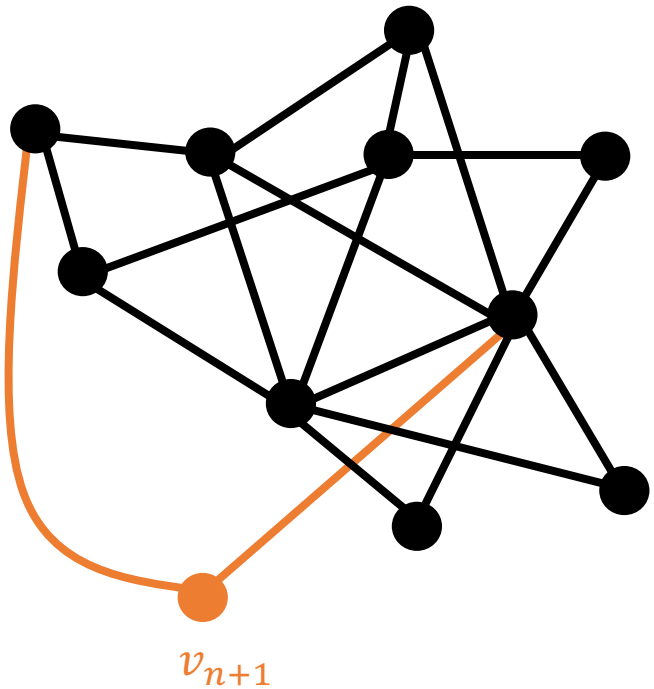

- Idea behind estimator: $(X_n^1(k) - X_n^2(k))^2 \asymp \frac{1}{t_\star}$
- **Coarse estimate** obtained by matching largest subtrees of the centroid

$$\frac{t_\star}{\log t_\star} \leq \hat{t} \leq t_\star \log t_\star$$

- **Finer estimate** obtained by averaging over the largest $\sim \log t_\star$ subtrees

# Future work

**Goal 1: Detection and estimation for multi-edge models**



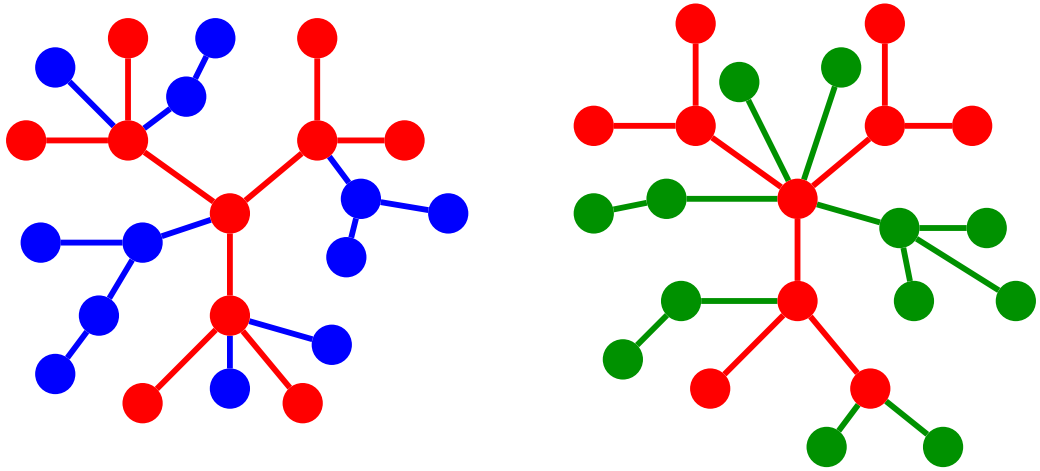

Examples:

- multi-edge UA/PA
- SI process on a non-tree base graph

- In multi-edge PA, could study max-degree vertices
- **Q:** How to generalize subtree centrality to non-trees?
  - Possibilities include eigenvector centrality
  - Implications for source/root estimation problems in general randomly growing graphs

# Future work

**Goal 2: Estimating the common subgraph**



**Q:** How well can we estimate the common subgraph $S_\star$?

- Conjecture: with high probability, any estimation algorithm will incorrectly estimate a constant fraction of vertices in $S_\star$
  - This is the case when given a single graph[1]. How much better can we do with two graphs?
- Possible approaches:
  - study properties of the $t_\star$ most central vertices
  - seeded matching approaches

1. G. Lugosi and A. S. Pereira. Finding the seed of uniform attachment trees. Electronic Journal of Probability, 24(18):1–15, 2019.